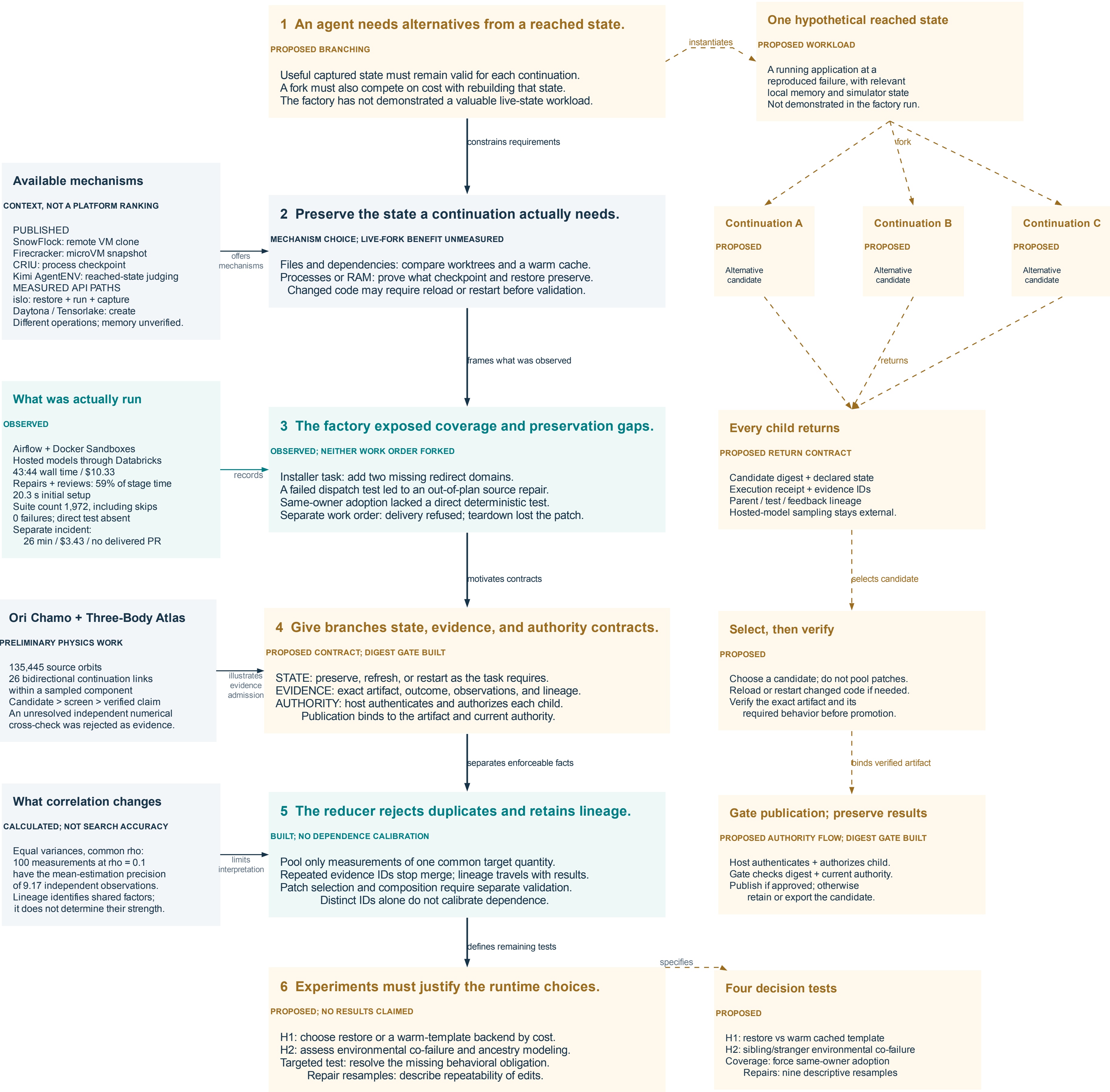


Forkable sandboxes need explicit state, evidence, and authority contracts

Useful state > observed gaps > enforceable contracts > measured decisions

Teal: observed or built Gold: proposed Navy: context or calculations



Arrows connect the seminar argument and design; they do not imply empirical causal proof. Case-study runs did not fork. Three-Body Atlas is a preliminary example of evidence admission, not a fork result. Continue on the two reading pages for mechanisms, precise observations, decision rules, and linked sources.

Useful state comes before the fork

Reading continuation - Acts 1 to 3. The poster is an argument map; these pages define what its boxes actually claim.

1 Alternatives need a task-relevant starting state.

PROPOSED

Consider a **hypothetical running application at a reproduced failure**, where useful local memory and simulator state are expensive to recreate. Forked continuations would be attractive only if that state survives, remains valid for their work, and costs less to capture, restore, and diverge than the alternatives. **The factory work orders did not demonstrate this live-state requirement.**

2 Choose the mechanism by what must survive.

PUBLISHED MECHANISMS; PROPOSED COMPARISON

Files and dependencies: compare independent worktrees and a warm cached template. **Processes and RAM:** check the actual snapshot semantics; a running process with a RAM-only nonce must survive restore into three children before calling the operation a memory fork. Changed source may require reload or restart. Hosted-model state and sampling remain external to the guest snapshot.

Context, not a ranking. SnowFlock, Firecracker, CRIU, and Kimi AgentENV provide different execution mechanisms. Exercised API paths were islo restore-run-capture and Daytona/Tensorlake creation. Those operations differ; memory capture in the islo measurements was unverified.

3 Two work orders expose concrete validation gaps.

OBSERVED

SELFHOST-2: add two installer redirect hosts across the policy and documentation. A four-replica dispatch race then failed: the cell-activation winner could differ from the valid lease holder. Repair added same-owner adoption on **CellBusy**, outside the six-file plan. The harness suite reported **1,972 including skips, zero failures**; the new adoption path lacked a direct deterministic test. Review approved with the gap still present.

Execution used Airflow, Docker Sandboxes, and models hosted through Databricks. **43:44 wall time; \$10.33.** Repairs and reviews consumed 59% of recorded stage time; initial setup took 20.3 seconds. **Neither work order forked.**

Separate preservation failure: a second work order was approved, but delivery refused the harness's unreviewed diff archive three times. Teardown then lost the patch: **26 minutes, \$3.43, no delivered PR.** Its raw records are unpublished; the public claim boundary documents the case.

Linked sources: [run metrics](#) | [repair diagnosis](#) | [final review](#) | [claim boundaries and runtime context](#).

Contracts turn branches into usable results

Reading continuation - Acts 4 to 6. Built enforcement, proposed authority, and unrun experiments remain separate.

4 Verify the candidate, then authorize publication.

PROPOSED CONTRACT; DIGEST GATE BUILT

State: declare preserved, refreshed, restarted, and external state. **Evidence:** bind checks and observations to the exact candidate and its lineage. **Authority:** the host authenticates and authorizes each child. The proposed pipeline **selects a candidate, verifies its exact artifact and required behavior, checks current authority, then publishes or retains/exports it**. Digest-bound approval is built; clone authority and epoch fencing remain proposed.

5 Duplicate rejection is useful but incomplete.

BUILT; NO DEPENDENCE CALIBRATION

The built reducer rejects a repeated evidence ID and carries lineage. Pooling applies only to measurements of **one common target quantity**. Selecting patches and verifying a composition are separate operations. Distinct IDs can still share state, tests, or feedback; lineage names those factors without estimating their effect.

Calculated illustration: with equal variances and common pairwise correlation $\rho = 0.1$, 100 measurements have the mean-estimation precision of **9.17 independent observations**. This is a variance model, not search accuracy, shared-bias correction, or a built confidence bound.

6 Four experiments lead to four different decisions.

PROPOSED; NO RESULTS CLAIMED

Experiment	Runtime decision
H1: restore vs warm cached template	Choose preparation backend by batch makespan and total cost; capture accounting must be explicit.
H2: siblings vs restored strangers	Model ancestry only if environmental co-failure supports it; this does not measure agent correctness.
Targeted adoption test	Resolve the behavioral obligation by forcing the branch; passing is not a fork speedup or accuracy result.
Nine repair resamples	Describe out-of-plan edits and distinct diffs; repeatability alone is not correlation or correctness.

Chamo / Three-Body Atlas: 135,445 source orbits; 26 bidirectional continuation links support one sampled component. Its candidate-to-screen-to-verified-claim workflow rejected an unresolved independent numerical cross-check. Preliminary physics work illustrates evidence admission; it does not establish a fork result.

Linked sources: [reducer paper](#) | [unrun experiment protocol](#) | [Chamo preprint](#) | [Atlas workflow](#).

Before running H2, amend inference for reused snapshot families and temporal dependence. The substitute public pre-repair state also lacks a confirmed tree-hash equivalence. These are unresolved design limits, not excluded results.